

MATH 213 – Homework 3

First: read the rest of chapter 1, and these parts of chapter 2: 2.1.1-2.1.3, the first paragraphs of 2.2 (about Experiments) and all of 2.3 (about Observational Studies). I have no formal questions on this reading, but it will help explain the terminology we've been looking at.

1. Elvis Presley is reported to have died in his Graceland mansion on August 16, 1977. On the 12th anniversary of this event, a Dallas record company wanted to learn the opinions of all adult Americans on the issue of whether Elvis was really dead. But of course they could not ask every adult American this question, so they sponsored a national call-in survey. Listeners of more than 100 radio stations were asked to call a 1-900 number (at a charge of \$2.50) to voice an opinion concerning whether Elvis was really dead. It turned out that 56% of the callers thought that Elvis was alive.
 - a. **What, do you think, is the population of interest (according to the producers at the radio station)?**
The population of interest is all adult Americans.
 - b. **Identify the sample for this study.**
The sample is the people who voluntarily call in to the 1-900 number to vote.
 - c. **What variable(s) were recorded? Classify each as categorical or numerical.**
 - 1) Belief about whether Elvis is alive or dead. (categorical)
 - 2) Willingness to call the 1-900 number and pay the fee. (categorical)
 - d. **What is the parameter of interest?**
True proportion of all adult Americans who believe Elvis is alive.
 - e. **What is the statistic of interest?**
The 56% of callers reported that they believed Elvis was alive.
- f. Do you think that the 56% accurately reflects the opinions of all Americans on this issue? (In other words, can the radio producers generalize the results of their survey to the population of all Americans?) If not, say whether you think 56% is too high or too low, identify some sources of sampling bias in the way the data were collected, and say what you would do differently if you were going to select a sample to try to answer this question.
 - 1) No, they cannot generalize the results to the population of all Americans.
 - 2) **Sampling bias:** the sample is biased because it only includes people who were motivated enough to call in. This is a voluntary response sample which typically overrepresents people with strong opinions.
 - 3) **Selection bias:** Many Americans may not have even known about the survey did not want to pay \$2.50 or just did not care for the question. This skews the sample towards those with higher interest in Elvis's theories.
 - 4) **56% too high or too low:** The 56% is too high because people who call are probably more likely to be Elvis fans and thus more likely to believe he is alive.

5) **Better sample/way to collect data:**

A better way to collect data would be to use a simple random sample of adult Americans instead of a voluntary response sample. For example, use a survey with free participation, this can still be done through phone calls or include more options such as email, or mailed question set.

2. Suppose that you want to estimate that average amount that students at Hood spent on textbooks during the 2021–2022 academic year. To do so, you gather data from a simple random sample of 50 Hood students. You can assume that all 50 students in our sample will happily give you this information.

a. For this research question identify

- **The population of interest:**
Students at Hood college
- **The variable of interest:**
Amount of money a student spent on a textbook during 2021 – 2022.
- **The parameter(s) of interest:**
The true average amount spent on textbooks by all Hood college students.
- **The associated statistic of interest:**
Sample mean of 50 students surveyed

b. Can we reasonably generalize our results to the population of all Hood students? Explain.

If the 50 students were selected using a simple random sample, then yes, we can generalize the results to the entire hood student population. However, if the sample was not random, for example if only students from a specific major were surveyed, then no, the results are not representative of the entire student population on campus.

Textbook problems:

Chapter 1: problem 10

Chapter 2: problem 12

CH1-P10)

10. **Efficacy of COVID-19 vaccine on adolescents, study components.** Results of a Phase 3 trial announced in March 2021 show that the Pfizer-BioNTech COVID-19 vaccine demonstrated 100% efficacy and robust antibody responses on 12 to 15 years old adolescents with or without prior evidence of SARS-CoV-2 infection. In this trial 2,260 adolescents were randomly assigned to two groups: one group got the vaccine ($n = 1,131$) and the other got a placebo ($n = 1,129$). While 18 cases of COVID-19 were observed in the placebo group, none were observed in the vaccine group.¹³ (Pfizer 2021)

a. **Is this an observational study or an experiment? Explain your reasoning.**

This type of study is an experiment because the researcher controlled the assignment of participants to group making this a controlled experiment rather than an observational study

b. What are the cases in this study and how many are included?

The cases in the study are the 2,260 adolescents who participated.

c. What is the response variable in the study and what type of variable is it?

The response variable is whether a participant developed covid-19 this is a categorical variable because it only has two outcomes (yes/no for developed covid-19)

d. What are the explanatory variables in the study and what types of variables are they?

The explanatory variable is whether a participant received the vaccine or the placebo, this is also a categorical variable as it only has two outcomes (received the vaccine or did not receive the vaccine)

CH2-P12)

12. **Stressed out.** A study that surveyed a random sample of otherwise healthy high school students found that they are more likely to get muscle cramps when they are stressed. The study also noted that students drink more coffee and sleep less when they are stressed.

a. What type of study is this?

This is an observational study because there was no manipulation of variables (no random assignment to stress or no stress groups)

b. Can this study be used to conclude a causal relationship between increased stress and muscle cramps?

No, this study cannot be used to conclude a causal relationship as observational studies can identify associations or correlations between variables, but they cannot establish causation.

c. State possible confounding variables that might explain the observed relationship between increased stress and muscle cramps.

- 1) Sleep deprivation
- 2) Increased coffee consumption
- 3) Physical activity levels
- 4) Dietary changes
- 5) Previous health conditions